Example 1.1

we substitute $y = Ce^{kt}$ into the LHS and RHS of the differential eqn and show that they are equal:

$$\text{LHS}: \frac{d}{dt}y = \frac{d}{dt}\left[Ce^{kt}\right] = Ce^{kt}\,k$$

$$\text{RHS}: ky = kCe^{kt}$$

Hence LHS = RHS i.e. y satisfies the ODE, $t \in (-\infty, \infty)$

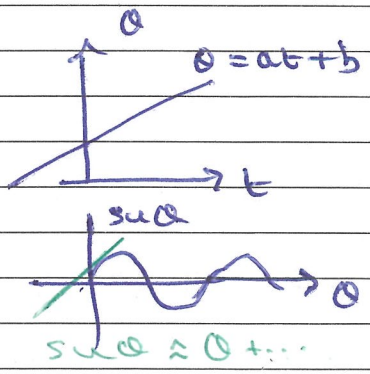
We have an explicit formula for the unknown $y(t)$
This is known as an explicit solution

Example 1.2

1. first
2. third

Example 1.3

1. linear
2. nonlinear due to $y\,dy$
3. nonlinear due to $\sin\theta$
4. linear



2/ First Order ODEs

linear ODE: $y' + p(t)y = g(t)$

Integrating factor $\mu(t) = e^{\int p(t)\,dt}$

$$\frac{d\mu}{dt} = e^{\int p\,dt}\,\frac{d}{dt}\left(\int p(t)\,dt\right) = p(t)e^{\int p\,dt}$$

$$\mu y' + p\mu y = \mu g$$

$$\frac{d}{dt}[\mu y] = \mu g$$

$$\mu\frac{dy}{dt} + y\frac{d\mu}{dt} = \mu g$$

$$\mu y' + y p \mu = \mu g$$

Example 2.1

$y' - 2y = e^{2t}$ $\quad y(0) = 2$

$y' + py = g$ where $p = -2$ $\quad g = e^{2t}$

Integrating factor $\mu = e^{\int p\,dt} = e^{\int -2\,dt} = e^{-2t}$

Multiply by μ:

$\mu y' - 2\mu y = \mu e^{2t}$

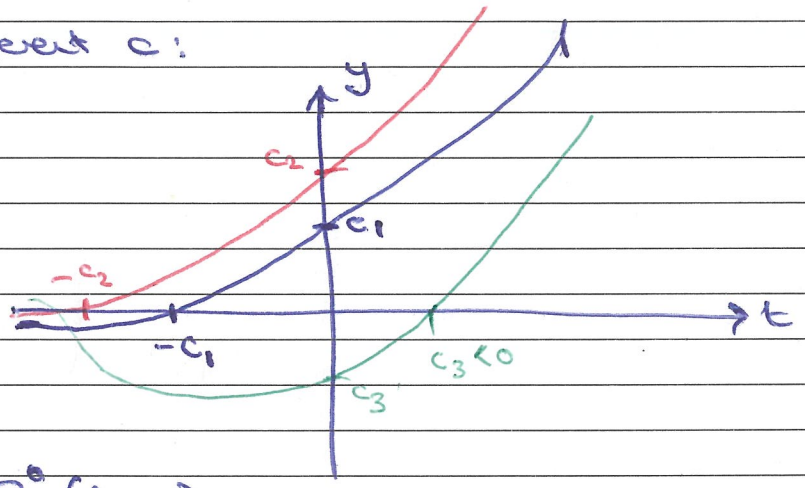
$\Rightarrow \frac{d}{dt}[\mu y] = \mu e^{2t}$ $\qquad \mu = e^{-2t}$

$\Rightarrow e^{-2t} y = \int e^{-2t} e^{2t}\,dt = \int 1\,dt = t + C$

$\Rightarrow y = e^{2t}(t + C)$ general solution, $t \in (-\infty, \infty)$

Integral curves

Plot for different C:



- @ $t = 0$ $\quad y = e^0(0 + C) = C$
- $y = 0$ $\quad \Rightarrow e^{2t}(t + C) = 0 \quad \Rightarrow t = -C$
- $t \to \infty$ $\quad y \to \infty$
- $t \to -\infty$ $\quad y \to 0^-$

IVP $\quad y(t=0) = 2$

$e^0(0 + C) = 2 \quad \Rightarrow C = 2 \quad \Rightarrow y = e^{2t}(t + 2), \; t \in (-\infty$